CSS Teaching Content

A comprehensive classroom guide to Cascading Style Sheets — from fundamentals to hands-on practice

BCA COURSE | BEGINNER LEVEL | LECTURE NOTES

What is CSS?

Definition, syntax, and the separation of concerns between HTML and CSS

Three Methods

Inline, internal, and external CSS — when to use each and why

Selectors & Properties

Colors, backgrounds, borders, padding, width/height, and the box model

Practice & Revision

Live demo, four hands-on tasks, and a complete quick-reference summary

What is CSS?

Definition & Role

CSS — **Cascading Style Sheets** — controls the visual presentation and layout of HTML elements on a webpage. While HTML defines *what* is on the page, CSS defines *how* it looks: color, font size, background, spacing, borders, width, height, layout, position, and responsive behavior.

This separation of concerns is a foundational principle of web development — keeping structure and style in distinct layers makes code cleaner and far easier to maintain.

What CSS Controls

- Color, font size, and typography
- Backgrounds, borders, and spacing
- Width, height, and layout
- Responsive design across screen sizes

The House Analogy

HTML	CSS	JavaScript
Structure of the house — walls, rooms, doors	Paint, decoration, size, spacing — how it looks	Electricity, automation, interaction — how it behaves

Basic Example

HTML creates content:

```
<h1>Hello World</h1>
```

CSS styles it:

```
h1 { color: blue; font-size: 40px; }
```

Why Do We Need CSS?

Without CSS

HTML alone renders everything in plain, unstyled browser defaults — no color, no layout, no visual hierarchy. Every heading looks the same. Every paragraph is black text on a white background. There is no concept of a button style, a card, or a centered layout. The page is functional but completely undesigned.

With CSS

The same HTML becomes visually appealing and professionally structured. Headings are centered and colored. Text is readable at the right size. Buttons have shape, color, and hover effects. Layouts respond to different screen widths. One well-written stylesheet transforms a raw document into a real webpage.

Five Core Advantages of CSS

1. **Separation of Design from Content**
   HTML handles structure; CSS handles appearance. Cleaner, more maintainable codebase.

2. **Write Once, Apply Everywhere**
   Reduces repeated styling — define a style rule once and it applies across every matching element.

3. **Responsive Design**
   CSS adapts layouts to different screen sizes — desktop, tablet, and mobile from a single file.

4. **Multi-page Control**
   One external stylesheet can control the visual design of an entire website — all pages updated instantly.

5. **Full Layout Control**
   Precise control over positioning, spacing, sizing, and visual hierarchy across every element.

CSS Syntax

The Basic Structure

```
selector { property: value; }
```

Every CSS rule follows this exact pattern. Each declaration ends with a semicolon `;`. Multiple properties can be listed inside a single rule block, one per line.

Three Parts Every Rule Must Have

- **Selector** — which element to target
- **Property** — what characteristic to change
- **Value** — what to set that characteristic to

Anatomy of a CSS Rule

Selector

Targets the HTML element(s) to style. Examples: `p`, `h1`, `.box`, `#title`

Property

The CSS attribute being changed. Examples: `color`, `font-size`, `background-color`

Value

The specific setting applied to the property. Examples: `red`, `20px`, `bold`

Example Rules

```
h1 {
  color: blue;
  background-color: yellow;
  font-size: 40px;
}
```

```
p {
 color: green;
 font-size: 18px;
 background-color: lightgray;
}
```

Three Ways to Add CSS

1

A. Inline CSS

Written directly inside the HTML element using the `style` attribute.

```
<h1 style="color: red;">Hello</h1>
```

Pros: Quick to write, useful for testing, targets one specific element with no extra files needed.
Cons: Hard to maintain at scale, mixes HTML and CSS in the same file, not suitable for large sites.

2

B. Internal CSS

Written inside a `<style>` tag placed in the `<head>` section of the HTML file. Best suited for styling a single webpage where no external file is needed.

```
<head>
  <style>
    h1 { color: blue; }
  </style>
</head>
```

3

C. External CSS

Written in a separate `.css` file and linked using a `<link>` tag. **Preferred for all real projects.**

```
<link rel="stylesheet"
href="style.css">
```

One CSS file can style multiple pages simultaneously: `index.html`, `about.html`, and `contact.html` all read from a single `style.css`.

External CSS is the industry standard. Always use it for projects with more than one page — it keeps your code DRY (Don't Repeat Yourself).

CSS Selectors

A selector tells CSS **which HTML elements** should receive the defined styles. Choosing the right selector is essential for writing precise, maintainable CSS.

Element Selector

Selects all elements of a given type.
`p { color: blue; }` — affects every `<p>` on the page.

Class Selector

Assigned to multiple elements using `class="name"`. CSS uses a dot prefix.
`.important { color: red; }`
Classes can be reused across different element types: `h1`, `p`, `div`.

ID Selector

Identifies one unique element using `id="name"`. CSS uses a hash prefix.
`#main-title { color: red; }`
Must be used only *once* per page.

Universal Selector

Selects *all* elements on the page.
`* { margin: 0; }`

Grouping Selector

Style multiple selectors at once with a comma-separated list.
`h1, h2, p { color: blue; }`

Class vs ID — Quick Comparison

Feature	Class	ID
Syntax	`.box`	`#box`
Reusable?	Yes — multiple elements	No — one unique element
Usage	Frequent, everyday styling	Unique elements only

Rule of thumb: default to classes. Use IDs only when an element is genuinely unique on the page — such as a page header or main navigation.

CSS Colors

CSS supports four primary methods for specifying color. Each has its own use case — from quick prototyping to pixel-precise design.

Color Name

Plain English names for common colors. Fast to write, limited palette.

```
color: red;
color: blue;
color: orange;
```

HEX Code

Six-digit hexadecimal value beginning with #. Most commonly used in professional design.

```
#ff0000 → Red
#00ff00 → Green
#0000ff → Blue
#000000 → Black
#ffffff → White
```

RGB

Specifies Red, Green, and Blue values from **0 to 255**.

```
color: rgb(255, 0, 0);
/* Red */
color: rgb(0, 0, 255);
/* Blue */
```

RGBA

Adds an **alpha (transparency)** channel as a fourth value. 0 = fully transparent; 1 = fully opaque.

```
color: rgba(255, 0, 0, 0.5);
/* 50% transparent red */
color: rgba(0, 0, 0, 0.8);
/* Dark semi-transparent */
```

ⓘ For beginners: start with color names for practice, then learn HEX codes — they are the standard used by design tools like Figma and Adobe XD.

Backgrounds

CSS provides a rich set of background properties that control both color fills and image-based backgrounds for any element.

Background Color

```
body {
  background-color: lightblue;
}
div {
  background-color: #eeeeee;
}
```

Applies a solid color fill to the entire element. Accepts color names, HEX, RGB, or RGBA values.

Background Image

```
body {
  background-image: url("background.jpg");
}
```

Loads an image file as the element's background. By default, the image tiles to fill the space.

Background Repeat

no-repeat

Displays the background image once only — no tiling.

repeat-x / repeat-y

Tiles the image horizontally or vertically only.

repeat (default)

Tiles the image in both directions to fill the element.

Background Size

cover

Image fills the entire area — may crop edges

contain

Entire image stays visible — may leave gaps

Borders & Border Radius

Basic Border Syntax

A border is defined with three values in sequence: **width, style, and color**.

```
border: 2px solid black;
```

Border Styles

solid

Continuous, unbroken line

dashed

Short dashes with gaps

dotted

A series of dots

double

Two parallel lines

Individual Side Borders

Target each side independently using specific properties:

```
border-top: 2px solid red;
border-right: 2px solid blue;
border-bottom: 2px solid green;
border-left: 2px solid orange;
```

Border Radius

Rounds the corners of an element's border box.

```
/* Rounded corners */
border-radius: 10px;

/* Perfect circle */
border-radius: 50%;
```

`border-radius: 50%` is commonly applied to images to make them circular — used in profile pictures and avatars.

Padding

Padding is the **space between the content and the border**. It pushes the border outward, making the element appear larger without adding external spacing.

Basic Padding

```
div { padding: 20px; }
/* Adds 20px on all four sides */
```

Individual Side Padding

```
padding-top: 10px;
padding-right: 20px;
padding-bottom: 15px;
padding-left: 5px;
```

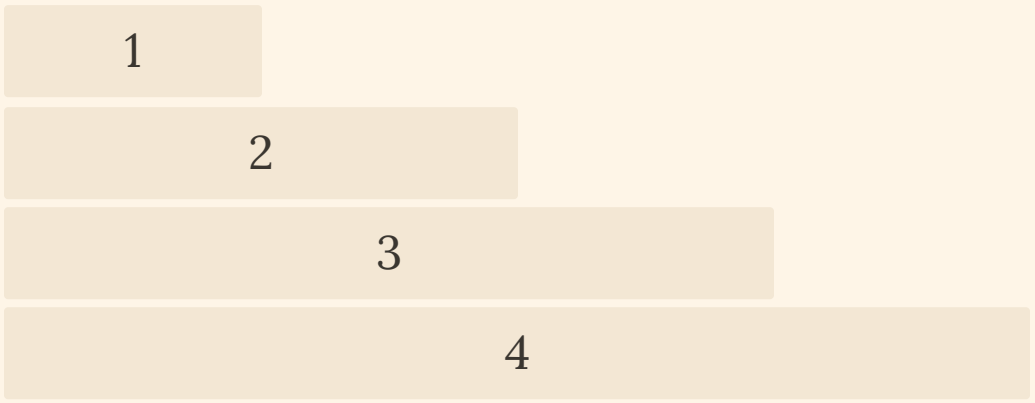
Shorthand Notation

```
/* Top/Bottom: 10px | Left/Right: 20px */
padding: 10px 20px;

/* Top → Right → Bottom → Left */
padding: 10px 20px 30px 40px;
```

Memory trick — TRBL (read as "Trouble"): Top, Right, Bottom, Left. This order applies to all four-value shorthand in CSS.

Visual Concept



1 **Margin**

Space outside the border — between elements

2 **Border**

Visible boundary around the element

3 **Padding**

Space between border and content

4 **Content**

The actual text or image inside

Increasing padding *pushes the border outward*. The content stays the same; the element grows.

Width, Height & CSS Units

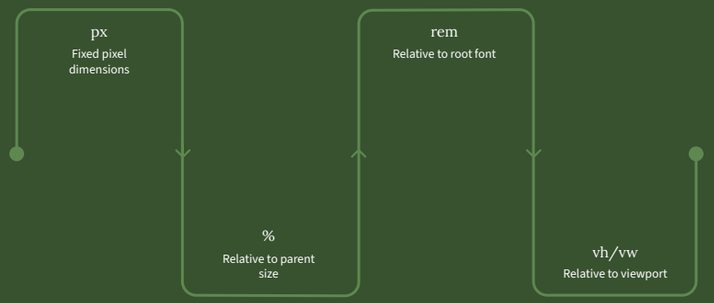
Width & Height

```
/* Fixed pixel dimensions */
width: 300px;
height: 200px;

/* Relative to parent */
width: 80%;
```

Fixed pixel values are rigid. Percentage values are fluid — they respond to the size of the parent container.

Beginner Learning Order



Common CSS Units Reference

Unit	Meaning	Example
`px`	Fixed pixel value — absolute	`width: 300px;`
`%`	Relative to parent element width	`width: 50%;`
`em`	Relative to element's/parent's font size	`padding: 2em;`
`rem`	Relative to root `<html>` font size	`font-size: 2rem;`
`vh`	1% of the viewport height	`height: 100vh;`
`vw`	1% of the viewport width	`width: 100vw;`

ⓘ `px` is the most intuitive unit for beginners. Move to `%` and `rem` once you're comfortable — they form the foundation of